

● EASY

● ALGEBRA

Linear equations in two variables

05

10 questions · ~15 min

Q1

ALGEBRA

x	y
0	18
1	13
2	8

The table shows three values of x and their corresponding values of y . There is a linear relationship between x and y . Which of the following equations represents this relationship?

- A. $y = 18x + 13$
- B. $y = 18x + 18$
- C. $y = -5x + 13$
- D. $y = -5x + 18$

A shipment consists of 5-pound boxes and 10-pound boxes with a total weight of 220 pounds. There are 13 10-pound boxes in the shipment. How many 5-pound boxes are in the shipment?

- A. 5
- B. 10
- C. 13
- D. 18

A total of 364 paper straws of equal length were used to construct two types of polygons: triangles and rectangles. The triangles and rectangles were constructed so that no two polygons had a common side. The equation $3x + 4y = 364$ represents this situation, where x is the number of triangles constructed and y is the number of rectangles constructed. What is the best interpretation of $x, y = 24, 73$ in this context?

- A. If 24 triangles were constructed, then 73 rectangles were constructed.
- B. If 24 triangles were constructed, then 73 paper straws were used.
- C. If 73 triangles were constructed, then 24 rectangles were constructed.
- D. If 73 triangles were constructed, then 24 paper straws were used.

For a camping trip a group bought x one-liter bottles of water and y three-liter bottles of water, for a total of 240 liters of water. Which equation represents this situation?

- A. $x + 3y = 240$
- B. $x + y = 240$
- C. $3x + 3y = 240$
- D. $3x + y = 240$

$$7x - 4y = -84$$

For the given equation, which table gives three values of x and their corresponding values of y ?

A.

x	0	4	8
y	21	28	35

B.

x	0	4	8
y	35	28	21

C.

x	21	28	35
y	0	4	8

D.

x	21	28	35
y	8	4	0

A food truck buys forks for \$ 0.04 each and plates for \$ 0.48 each. The total cost of x forks and y plates is \$ 661.76. Which equation represents this situation?

A. $0.48x - 0.04y = 661.76$

B. $0.04x - 0.48y = 661.76$

C. $0.48x + 0.04y = 661.76$

D. $0.04x + 0.48y = 661.76$

Q7

GRID-IN ALGEBRA

The y -intercept of the graph of $12x + 2y = 18$ in the xy -plane is $0, y$. What is the value of y ?

STUDENT-PRODUCED RESPONSE

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0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

$$4x + 3y = 24$$

Mario purchased 4 binders that cost x dollars each and 3 notebooks that cost y dollars each. If the given equation represents this situation, which of the following is the best interpretation of 24 in this context?

- A. The total cost, in dollars, for all binders purchased
- B. The total cost, in dollars, for all notebooks purchased
- C. The total cost, in dollars, for all binders and notebooks purchased
- D. The difference in the total cost, in dollars, between the number of binders and notebooks purchased

Line k is defined by $y = 6x + 4$. Line j is parallel to line k in the xy -plane and passes through the point $(0, 5)$. Which equation defines line j ?

- A. $y = 6x + 5$
- B. $y = -5x + 5$
- C. $y = -6x + 5$
- D. $y = 5x + 5$

Tony spends \$80 per month on public transportation. A 10-ride pass costs \$12.50, and a single-ride pass costs \$1.50. If g represents the number of 10-ride passes Tony buys in a month and t represents the number of single-ride passes Tony buys in a month, which of the following equations best represents the relationship between g and t ?

- A. $g + t = 80$
- B. $g + t = 1.50 + 12.50$
- C. $1.50g + 12.50t = 80$
- D. $12.50g + 1.50t = 80$